

— SWIPE TO READ

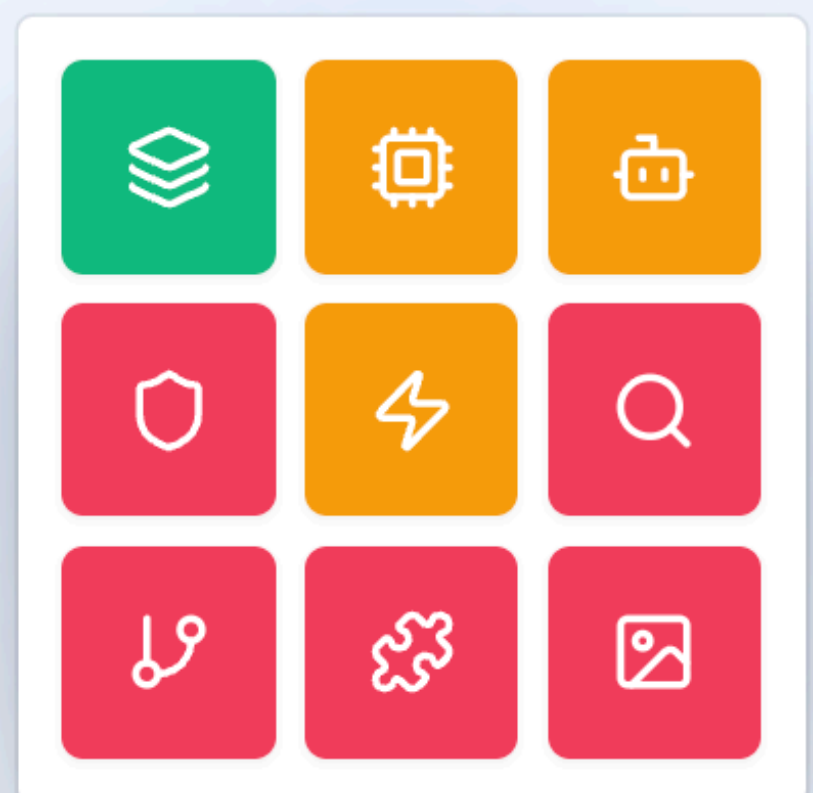
Nine RAG patterns, *compared.*



From beginner **Standard RAG** to advanced **Multimodal** — when to use which.



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What is RAG ?

RAG (Retrieval-Augmented Generation) combines information retrieval with LLMs to answer questions using your own documents. Instead of relying on training data, RAG retrieves relevant context from your documents and uses it to generate accurate answers.

WHY RAG?

- Answer questions from your own documents
- No need to retrain models
- Reduces hallucinations by grounding in context
- Works with any LLM and document type



Simple RAG

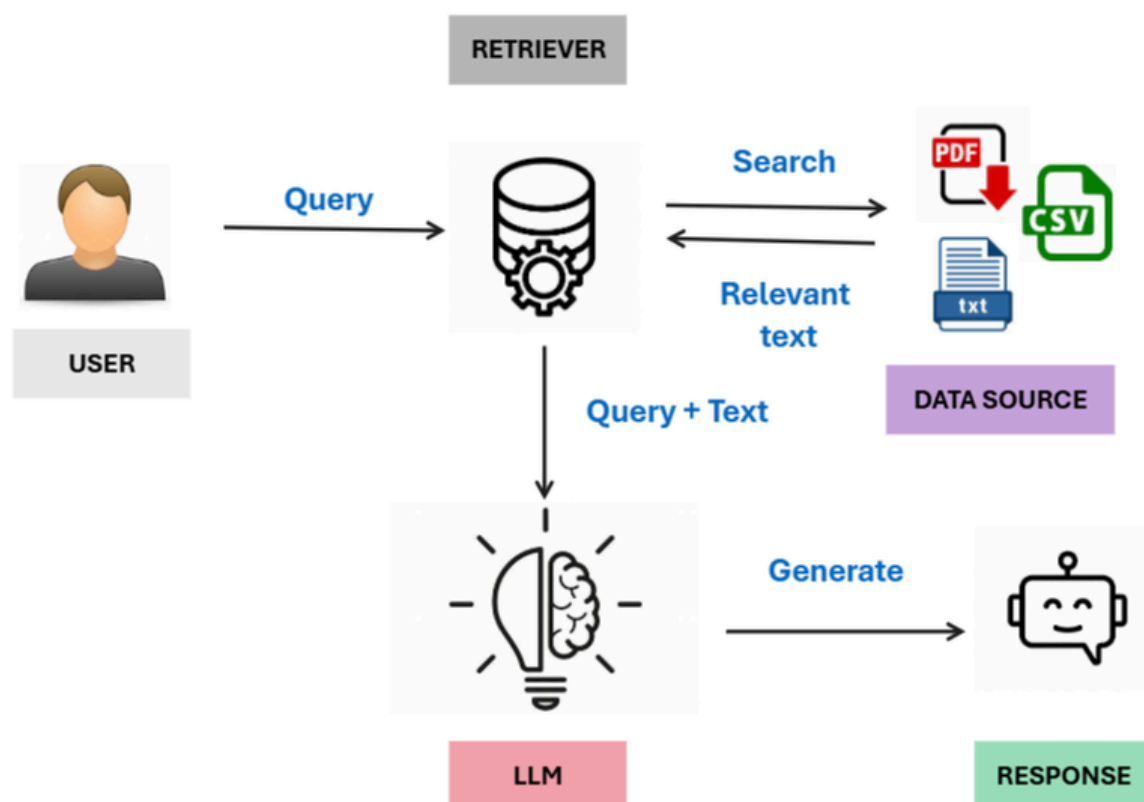
The foundation of all RAG systems. User query → Retriever → Top-K documents fetched → LLM generates answer using context.

BEST FOR:

- ✓ FAQs
- ✓ Knowledge base search
- ✓ Basic chatbots

⚠ LIMITATION:

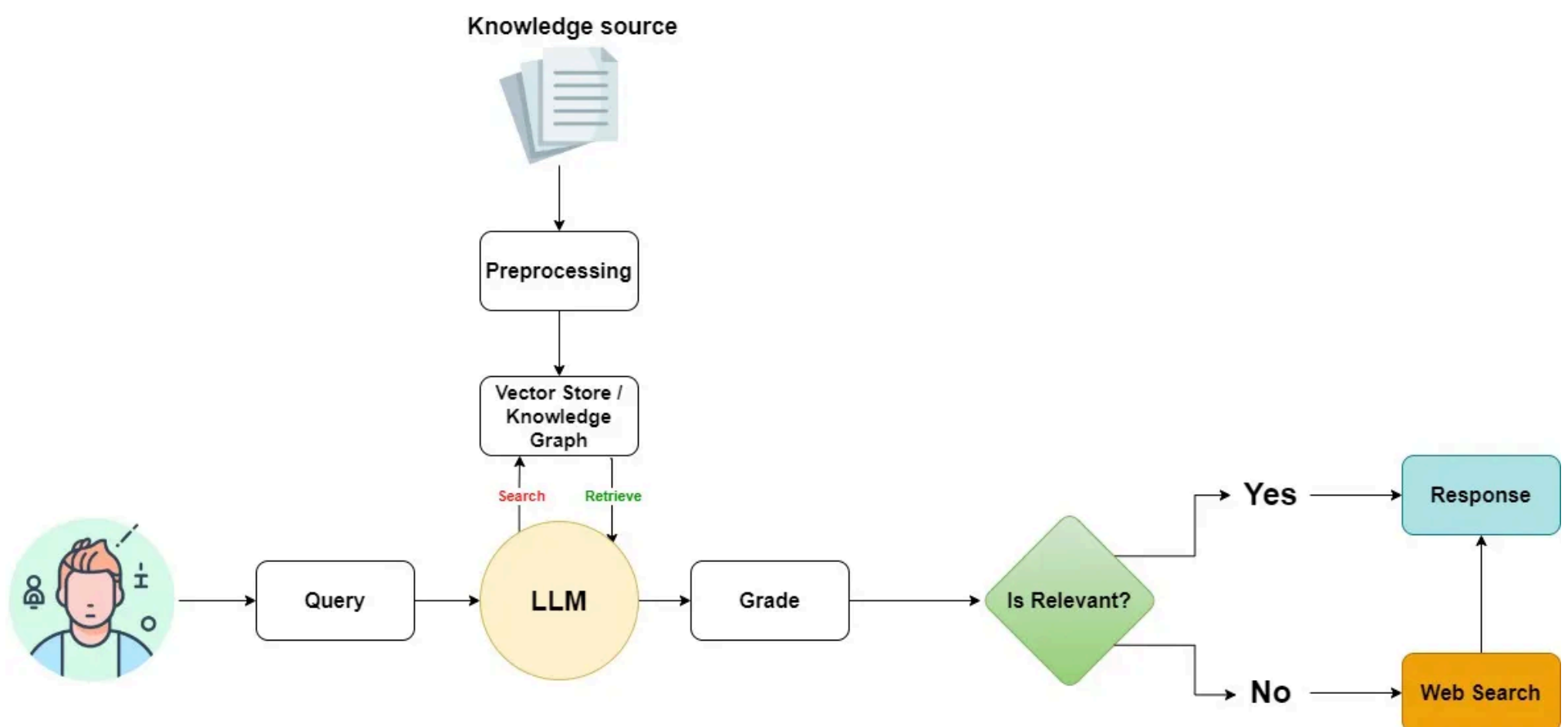
No validation if the answer is wrong



RAG with validation & correction. Retrieved content is graded, compared against trusted rules or datasets. If inaccurate → system re-searches (web or KB) → Answer is corrected automatically.

BEST FOR:

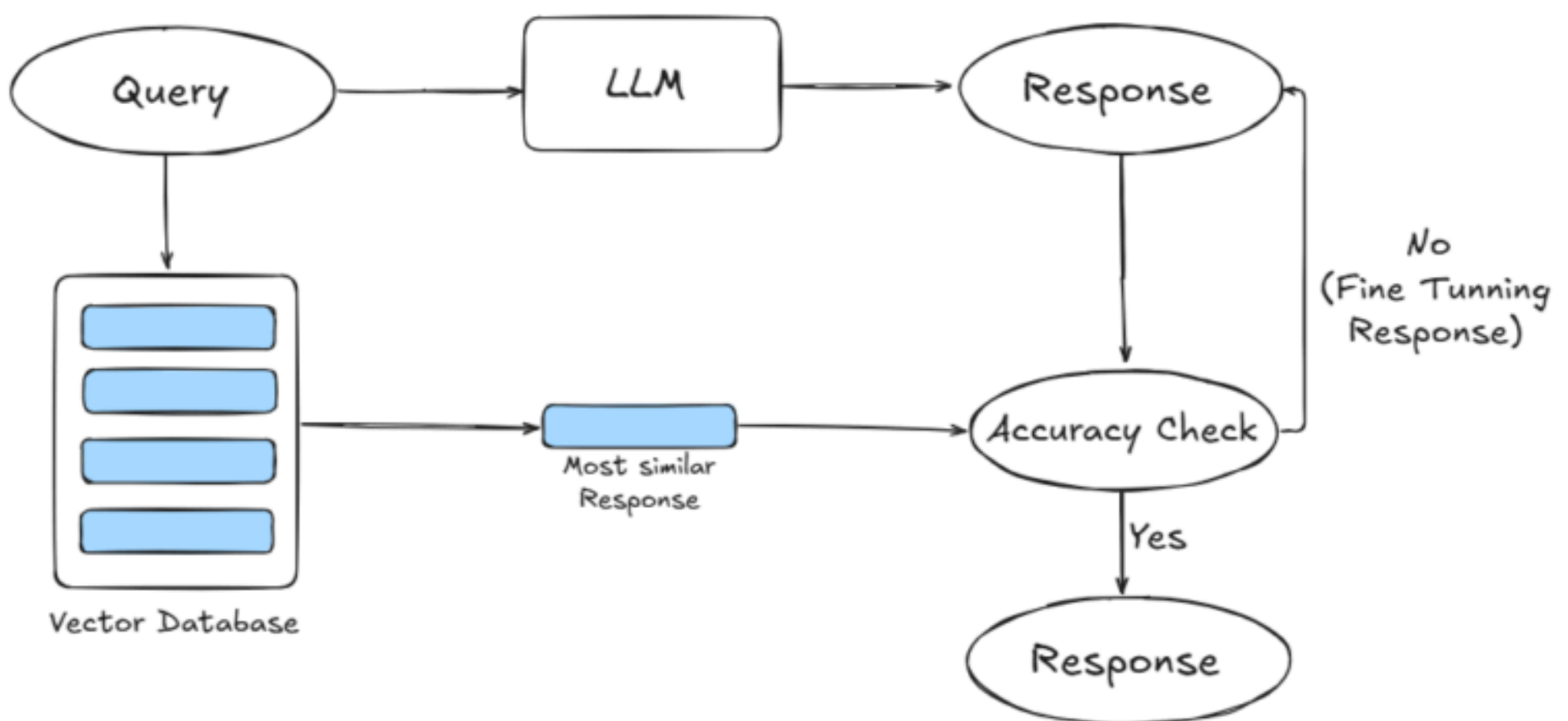
- ✓ Compliance systems
- ✓ Enterprise QA
- ✓ High-accuracy applications



RAG that reflects on itself. The LLM critiques its own answer. Process: Retrieve → Generate → Check relevance → Detect hallucinations → Improve answer iteratively.

BEST FOR:

- ✓ Research assistants
- ✓ Analytical tasks
- ✓ Complex reasoning workflows



Speculative RAG

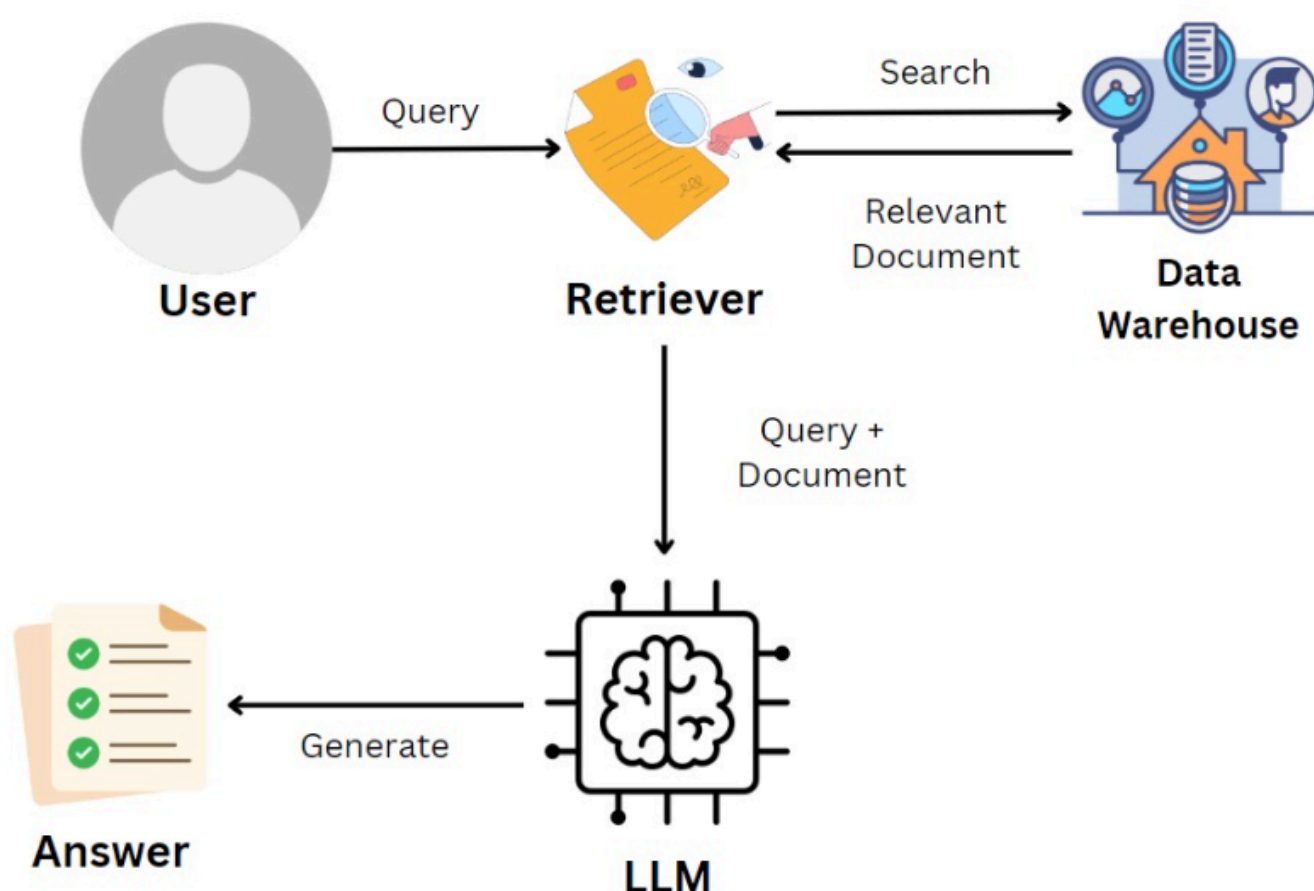
Multiple answers → pick the best one. Instead of one response, LLM generates multiple candidate answers. Each answer is graded. Best-scoring response is selected.

WHY IT MATTERS:

Handles ambiguity, improves answer quality, reduces single-path failure

USED IN:

- Open-ended questions
- Exploratory search
- Decision support systems

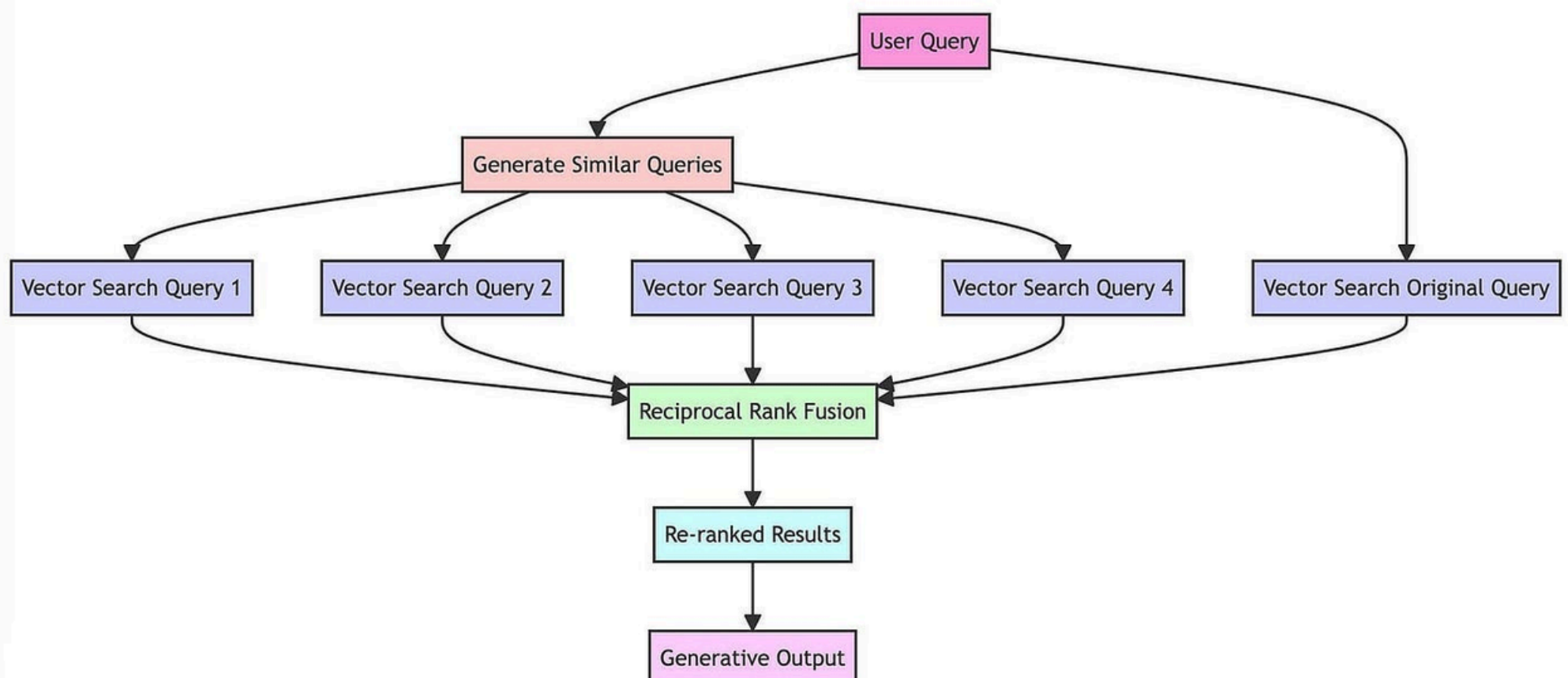


Combining knowledge from multiple sources. Retrieves documents from different perspectives, merges consistent information, resolves conflicts logically, produces a balanced final answer.

BEST FOR:

- ✓ Research synthesis
- ✓ Policy analysis
- ✓ Multi-source reporting

Think: "LLM as an intelligent editor"

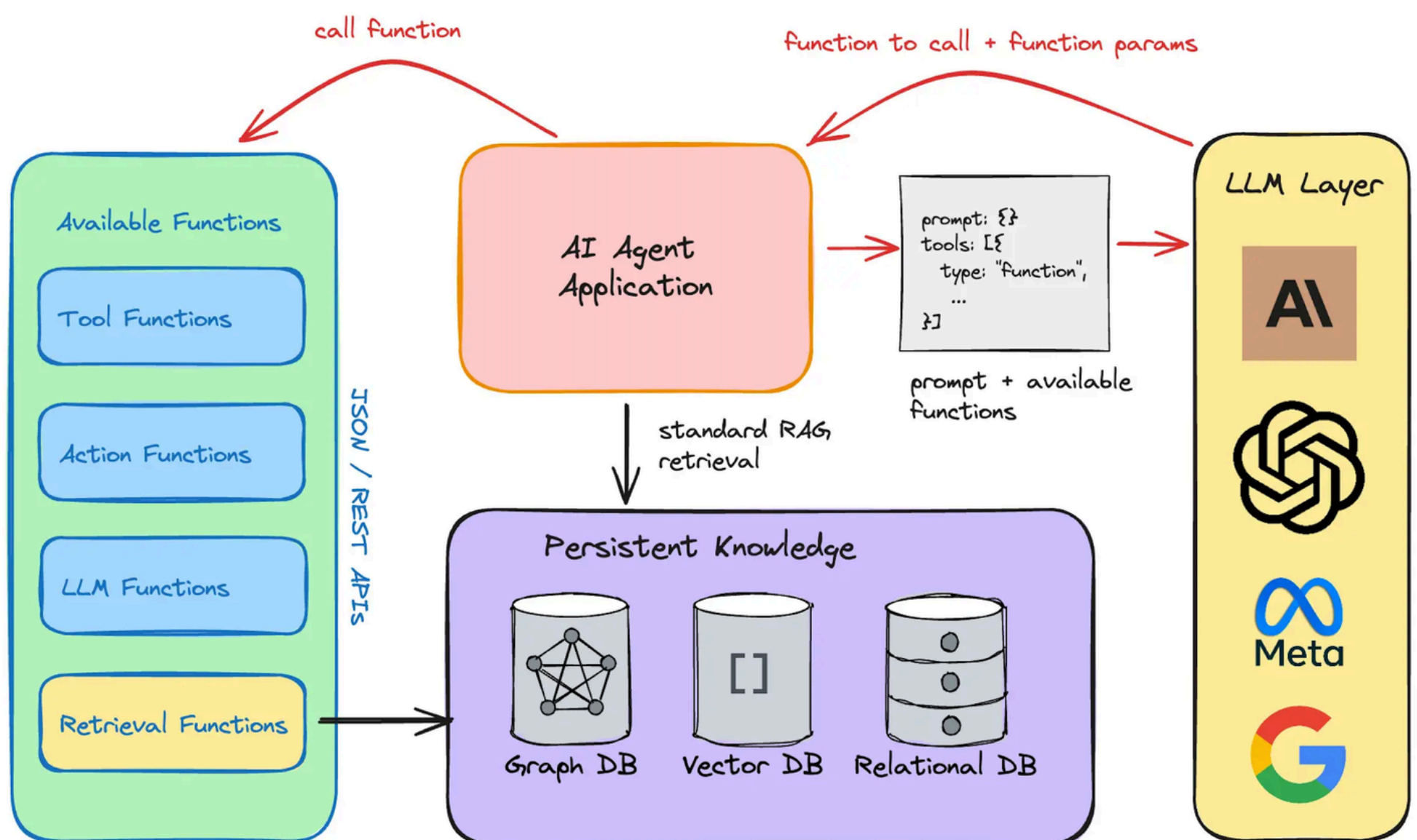


Agentic RAG

RAG that can think, decide, and act. Checks if the answer is sufficient. If not → takes action. Uses tools, searches again, re-plans. Loops until goal is achieved.

BEST FOR:

- ✓ AI agents
- ✓ Autonomous workflows
- ✓ Multi-step problem solving



Which RAG Should You Use?

Quick guidance 📌

- **Simple RAG** → Basic QA
- **Corrective RAG** → Accuracy-critical apps
- **Self RAG** → Reasoning & research
- **Speculative RAG** → Ambiguous queries
- **Fusion RAG** → Multi-source knowledge
- **Agentic RAG** → Autonomous AI systems

👉 Choose based on **problem complexity**, not hype.



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